Lecture 8 – Conductors Continued

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- Key properties of conductors & justification.
- Using conductors in problems.

Example 1: Gaussian surface

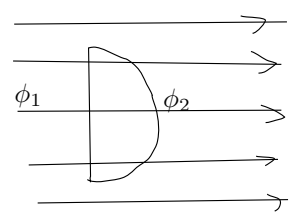

Figure 1: Conceptual example 4

ϕ_1 is negative, as field lines lead into surface.
ϕ_2 is positive, as field lines lead out of the surface.

Example 2: Cylindrical shell

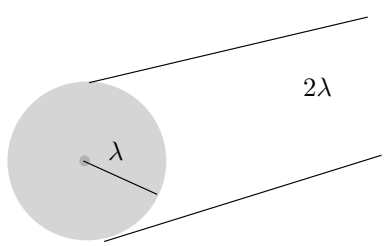

Figure 2: Problem 49

A thin hollow cylindrical shell with charge density 2λ surrounds a line charge λ as shown.
What is $\vec{E}$ at $r < R, r > R$?

$r < R$:

$$\oiint \vec{E}.d\vec{A} = \frac{Q_{in}}{\epsilon_0}$$

$$EA = \frac{\lambda L}{\epsilon_0}$$

$$E2\pi rL = \frac{\lambda L}{\epsilon_0}$$

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

$r > R$:

$$\oiint \vec{E}.d\vec{A} = \frac{Q_{in}}{\epsilon_0}$$

$$EA =$$

$$E2\pi rL = \frac{\lambda L + 2\lambda L}{\epsilon_0}$$

$$E = \frac{3\lambda}{2\pi\epsilon_0 r}$$

Conductors – Key Points and Justification:

1. $\vec{E} = 0$ inside of conductors.

$$\vec{F} = q\vec{E}$$

   (a) Conductor will become electrically charged in the direction of the electric field.

$$\vec{E_{tot}} = \vec{E_{ext}} + \vec{E_{ind}} = 0$$

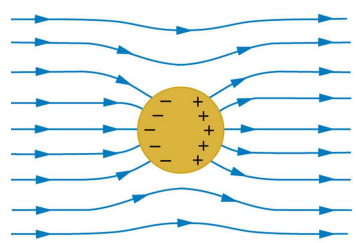

Figure 3: Conductor in an electric field.

2. Any net or excess carge lies only on the surface of conductors.

$$\oiint \vec{E}.d\vec{A} = \frac{Q_{in}}{\epsilon_0}$$

$$0 = \frac{Q_{in}}{\epsilon_0}$$

$$Q_{in} = 0$$

3. The electric field at the surface of a conductor is $\frac{\eta}{\epsilon_0}$.

$$E = \frac{\eta}{\epsilon_0}$$

4. Electric field is zero inside any cavity in the conductor, unless there is a charge inside of said cavity.

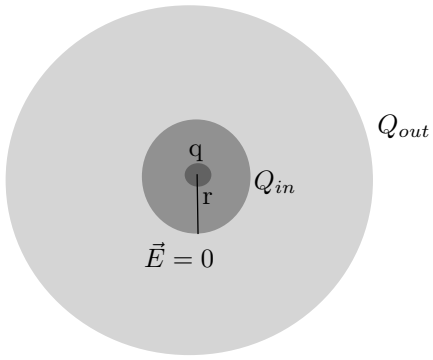

Figure 4: Cavity example

$$\oiint \vec{E}.d\vec{A} = \frac{Q_{in}}{\epsilon_0}$$

$$0 = \frac{q + q_{in}}{\epsilon_0}$$

$$q_{in} = -q$$